# Reddit Hyperlink Network:
# Conflict, Connectivity, and Community Structure

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## Introduction

Born in 2005, Reddit has grown into one of the largest and most influential platforms for online discussion. What makes Reddit unique is its structure. Instead of functioning as a single community, it is made up of thousands of subreddits, each acting as its own space with its own norms, interests, and culture. At the same time, these communities are not isolated. They constantly interact through links, references, and shared discussions, forming a much larger interconnected network.

While many of these interactions are neutral or positive, not all of them are cooperative. Conflict between communities is also a visible and recurring part of Reddit. Subreddits call each other out, criticize content, and sometimes engage in ongoing disputes. Such interactions raise an important question: do they simply reflect disagreement, or do they actually shape how the network itself is structured?

The goal of this project is to understand how conflict between subreddits shapes Reddit's overall network structure. Rather than treating conflict as isolated behavior, this analysis considers it as part of a larger system and examines how it fits into the broader pattern of interactions across the platform. To guide this analysis, the following sub questions were explored:

- ❖ Do negative links increase separation between groups?
- ❖ Are hostile interactions more likely to occur between communities?
- ❖ Which subreddits act as bridges in the network?
- ❖ Do some subreddits attract more negative attention?
- ❖ Does the structure of the network change over time?

The dataset used for this project is the Reddit Hyperlink Network from Stanford's SNAP database and covers interactions from January 2014 to April 2017. In this network, nodes represent subreddits, and edges represent directed hyperlinks from one subreddit to another. Each edge also includes a sentiment label, either positive or negative. This allows the network to be analyzed as a signed system where both cooperative and conflicting interactions contribute to its overall structure.

## Data Collection and Preprocessing

The dataset contains more than 286,000 total links. Before performing any network analysis, several preprocessing steps were applied to ensure consistency and usability.

First, in order to deal with Timestamp, the column was converted into a datetime format. This transformation made it possible to analyze how the network evolves over time and to split the data into different time periods.

Next, the properties column was expanded into 86 separate numerical features. These were originally stored as a single comma-separated string and converting them into structured columns helped ensure that the dataset was fully numeric and clean.

A complete missing value check was also performed, and it confirmed that there were no missing values.

Edges were then aggregated so that each unique source and target pair formed a single connection. This step simplified the network and removed redundancy.

After preprocessing, the final network consisted of:

- 35,776 nodes
- 137,821 directed edges

In terms of sentiment distribution:

- 93% links were positive or neutral
- 7% links were negative

This imbalance is important to note because negative interactions make up a much smaller portion of the network, meaning positive interactions are likely to dominate the overall structure. However, the negative subset is still large enough to be analyzed as its own network.

Additionally, based on the EDA, a small number of subreddits drive most of the network activity. On the source side, subredditdrama , r/circlebroke, and r/shitliberalssay act as the main broadcasters, sending content out across the network. But on the target side, askreddit (7,329 incoming links) is the biggest hub, followed by iama and pics, which receive most of the incoming traffic.

## Network Analysis

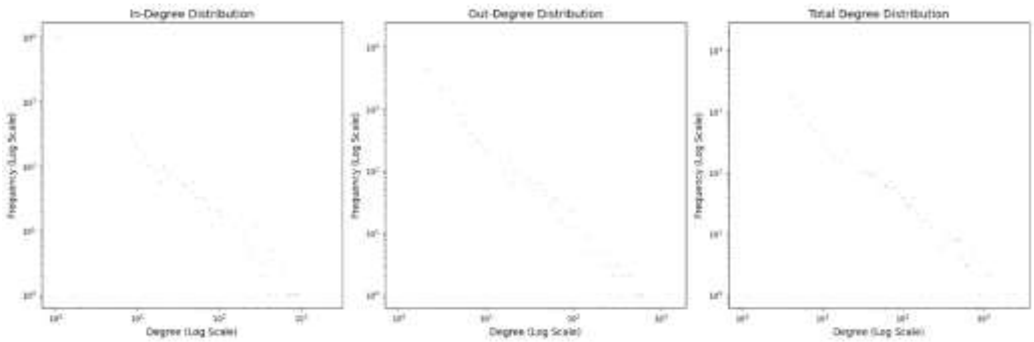


Figure 1.Degree Distribution

All 3 degree distributions are very skewed. Most subreddits have only a few connections, while a small number have a very large number of connections. A few hubs like askreddit, iama, pics, videos dominate the network, while most communities sit on the edges with little activity. This shows a hub-based (scale-free–like) structure, where a small group of subreddits controls most of the connectivity in the network.

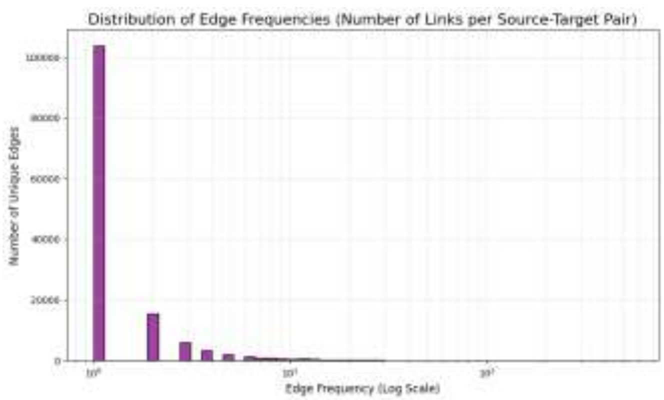


Figure 2. Edge Frequency

This figure shows how often subreddit pairs interact. The distribution is highly skewed: most pairs appear only once, while a few repeat frequently, with a maximum of 548 interactions. This suggests Reddit is mostly made up of one-off connections, with limited repeated interaction between communities. However, a small number of subreddit pairs form strong, persistent links through repeated engagement.

Overall, this distribution shows that Reddit is mostly a sparse interaction network with occasional dense interaction pockets. These pockets are important because they highlight where relationships are sustained over time, and where repeated attention or conflict is likely to emerge.

## Connected Components

| Metric | Value |
|---|---|
| Total SCCs | 24,071 |
| Largest SCC Size | 11,564 nodes |
| Total WCCs | 497 |
| Largest WCC Size | 34,671 nodes |
| Nodes in Largest WCC (Undirected) | 34,671 |
| Edges in Largest WCC (Undirected) | 123,570 |
| Global Clustering Coefficient | 0.1862 |
| Graph Density | 0.0002 |
| Average Degree (Largest WCC) | 7.1281 |
| Estimated Avg Shortest Path Length | 3.9665 |
| Estimated Diameter | 11 |

Table 1.Connected Components

The structure of connectivity was then analyzed using both strongly connected and weakly connected components. The network contains 24,071 strongly connected components (SCCs) and 497 weakly connected components (WCCs), showing a clear difference between directed and undirected structure.

In the directed view, the network is highly fragmented because most subreddit links are one-way, so communities are often not mutually reachable. But when direction is ignored, almost the entire network becomes connected, with the largest component including 34,671 of 35,776 nodes. This shows Reddit functions as one large interconnected system, even if relationships are not reciprocal.

Despite being very sparse, density = 0.0002, the network still shows meaningful local structure, with a clustering coefficient of 0.1862. This indicates that subreddits tend to form tight community groups where connections cluster rather than occur randomly.

Globally, the network is highly efficient. The average shortest path length is about 3.97, and the diameter is 11, meaning any subreddit can be reached from another in only a few steps. This reflects a small-world structure, where hubs help keep the network tightly connected.

Overall, Reddit is globally connected but directionally fragmented and locally clustered. Essentially, a small number of hubs hold the system together and enable fast flow across an otherwise sparse and uneven network.

## Centrality Analysis

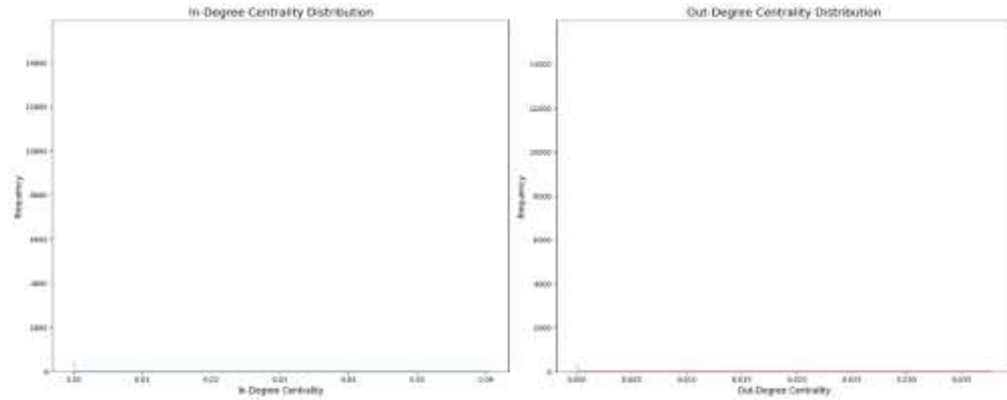


Figure 3. In degree vs Out degree

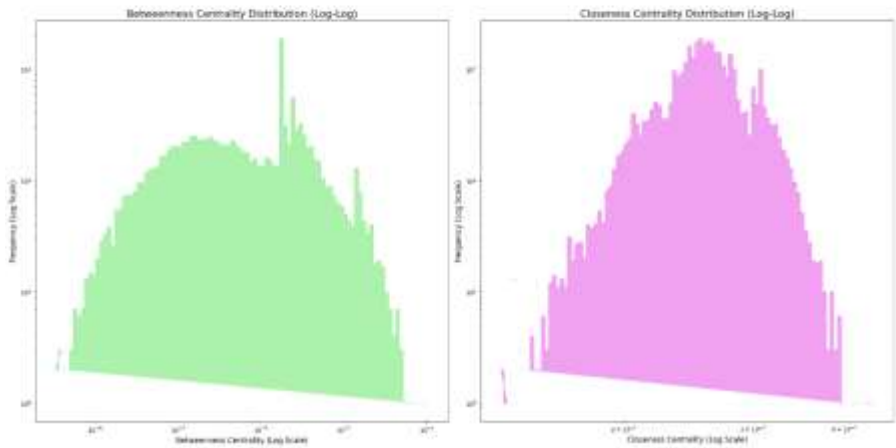


Figure 4. Edge Frequency

| Centrality Type | What It Measures | Top Subreddits (with representative values) |
|---|---|---|
| **In-Degree Centrality** | Subreddits most *linked to* | **askreddit (0.0604)**, iama (0.0460), pics (0.0266), videos (0.0246), todayilearned (0.0228) |
| **Out-Degree Centrality** | Subreddits that *link out* the most | **subredditdrama (0.0377)**, copypasta (0.0178), drama (0.0168), subredditoftheday (0.0156), outoftheloop (0.0142) |
| **Betweenness Centrality** | Subreddits acting as *bridges* between communities | **askreddit (0.1163)**, iama (0.0845), subredditdrama (0.0761), writingprompts (0.0505), outoftheloop (0.0361) |
| **Closeness Centrality** | Subreddits closest to all others | **askreddit (0.4398)**, iama (0.4296), subredditdrama (0.4294), outoftheloop (0.4122), videos (0.3994) |
| **Eigenvector Centrality** | Subreddits connected to other *important* subreddits | **askreddit (0.1998)**, subredditdrama (0.1725), iama (0.1684), outoftheloop (0.1285), pics (0.1151) |
| **PageRank** | Influence based on incoming links from important nodes | **askreddit (0.00813)**, iama (0.00627), subredditdrama (0.00493), writingprompts (0.00489), tipofmypenis (0.00384) |

Table 2 .Centrality Findings

Centrality measures are often used to understand different types of importance in the network. In simple terms, each one looks at the network from a slightly different angle. For example, in-degree captures how much attention a subreddit receives, out-degree shows how much it links out to others, betweenness identifies subreddits that sit "in between" others and connect different parts of the network, closeness measures how quickly a node can reach all others, eigenvector centrality captures whether a node is connected to other important nodes, and PageRank measures overall influence based on being linked by other influential subreddits.

Across all of these measures, the same pattern consistently appears: most subreddits score very low, while a small number dominate across nearly every metric. This already tells us the network is highly uneven and structured around a few key hubs rather than being evenly distributed.

In terms of in-degree centrality, subreddits like *askreddit, iama, pics, videos, and todayilearned* receive the most incoming links. This means they act as major attention sinks in the network; many other subreddits point toward them, so they function as shared reference points across Reddit.

Out-degree centrality shows a different behavior. Communities like *subredditdrama, copypasta, and outoftheloop* are the most active in linking outward. These are more "meta" or discussion-driven spaces that constantly reference other subreddits rather than just receiving attention.

Betweenness centrality highlights a more structural role. Subreddits such as *askreddit, iama, and subredditdrama* appear frequently on shortest paths between other nodes, meaning they connect otherwise separate parts of the network. This is important because it means they don't just participate in discussions, they control how easily content and interaction can move between different communities.

Closeness centrality reinforces this idea of central positioning. Again, *askreddit and iama* stand out because they can reach all other subreddits in very few steps. In practical terms, this just means they are centrally embedded in the network and sit close to everything else, making them ideal starting points for information spread.

Eigenvector centrality adds another layer by focusing not just on how many connections a node has, but on who those connections are with. The same core subreddits score highly here, meaning they are not just connected broadly, but are also connected to other well-connected and influential nodes.

Finally, PageRank confirms overall influence in a directed sense. Subreddits like *askreddit, iama, and subredditdrama* again rank the highest, showing that they receive links not just frequently, but from other important subreddits. This strengthens their role as persistent, high-visibility nodes in the network.

Overall, all centrality measures reinforce the same structure: Reddit is organized around a small set of highly influential hubs. These hubs act as attention centers, bridges between communities, and core influencers. Rather than importance being spread across many communities, it is concentrated in a few subreddits that effectively shape how information, discussion, and interaction flow through the entire network.

## Comparison with ER and BA Models

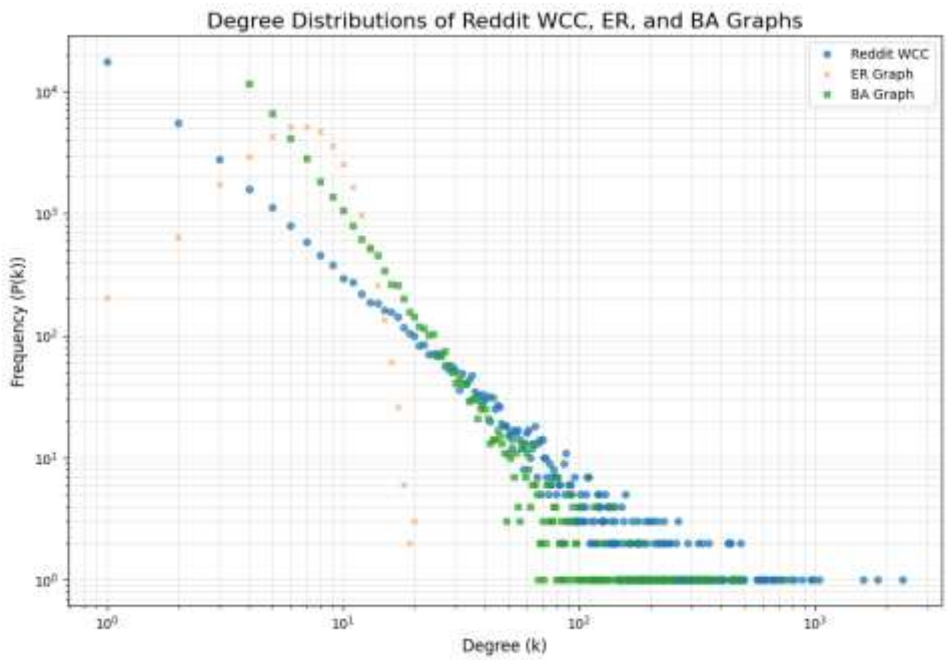


Figure 5. In degree vs Out degree

| Network Model | Clustering Coefficient | Avg. Shortest Path Length | Structural Characteristics |
|---|---|---|---|
| **Erdős–Rényi (ER)** | **0.0002** | **5.56** | Mostly random structure; no hubs; very low clustering |
| **Barabási–Albert (BA)** | **0.0023** | **4.30** | Scale-free with hubs; slightly higher clustering but still low |
| **Real Reddit Network** | **0.1862** | **~3.87** | Strong community structure; highly clustered; short paths; hub-dominated |

Table 3. ER vs BA Comparison

To better understand the structure, I compared Reddit network against the standard graph models: the Erdős–Rényi (ER) model, which is random, and the Barabási–Albert (BA) model, which produces hub-based networks through preferential attachment.

As shown in Table 3, the ER model has very low clustering (0.0002) and long average path lengths (5.56), meaning connections form randomly with no meaningful community structure. The BA model introduces hubs and improves efficiency, giving shorter paths (4.30), but it still has very low clustering (0.0023), so it does not produce strong local grouping or community structure.

The Reddit network, however is different from both. It combines high clustering (0.1862) with short path lengths (~3.87), while also maintaining a strong hub-based structure. This means the network is not only efficient at a global level like BA, but also highly organized into tightly connected communities, which neither ER nor BA captures well on their own.

Overall, Reddit behaves partially like a BA network due to its hubs, but its much higher clustering shows that real structure is driven just as much by community formation as by preferential attachment.

## Assortative Mixing

| Metric | Value |
|---|---|
| **Degree Assortativity** | **-0.0930** |
| **Community Assortativity** | **0.5526** |

Table 4. Assortative Mixing

Assortative mixing measures homophily, i.e whether nodes tend to connect to others that are similar to them. Here, similarity is defined in two ways: degree assortativity- which is based on how connected a node is and community assortativity- which based on community membership.

The degree assortativity is –0.0930, which indicates a disassortative structure. In practice, this means high-degree subreddits tend to connect with low-degree subreddits, rather than with other hubs. Essentially, large communities like AskReddit or IAmA attract links from many smaller subreddits, but they do not primarily link to each other. As a result, connectivity flows through hubs rather than forming an interconnected elite core.

In contrast, community assortativity is 0.5526, which is strongly positive. This shows that subreddits are much more likely to connect within their own communities than across them.

Together, these results show a dual structure. On one hand, hubs pull connections in from across the network (disassortative degree mixing), while on the other hand, most activity still stays inside tightly defined communities (strong community assortativity). This combination explains why Reddit is both globally connected and highly clustered.

## Time Based Evolution of Graph

| Metric | Early Period | Late Period |
|---|---|---|
| **Nodes** | 16,781 | 28,469 |
| **Edges** | 53,280 | 98,308 |
| **Nodes (Undirected WCC)** | 16,781 | 28,469 |
| **Edges (Undirected WCC)** | 48,454 | 89,216 |
| **Clustering Coefficient** | 0.1564 | 0.1648 |
| **Graph Density** | 0.000344 | 0.000220 |
| **Avg Shortest Path Length** | 4.0023 | 4.0284 |

Table 5. Time Based Evolution

To understand how Reddit's structure changes over time, the network was split into two periods: before 2015-06-01 (early) and after 2015-06-01 (late). This allowed us to see whether growth changes the way subreddits connect.

As you can see in Table 5, the network grows substantially over time. Nodes increase from 16,000 to 28,000, and edges rise from 53,000 to 98,00, showing a large expansion in both the number of communities and cross-subreddit interactions. Despite this growth, the largest connected component remains dominant in both periods, meaning Reddit stays globally connected as it scales.

Structurally, the network becomes slightly more locally clustered over time, with the clustering coefficient increasing from 0.1564 to 0.1648. This suggests that as Reddit grows, subreddits tend to form tighter community groupings rather than spreading connections evenly. At the same time, density decreases from 0.000344 to 0.000220, which is expected in large growing networks where new nodes appear faster than new links form, making the overall structure sparser.

Despite these changes, however, global connectivity remains stable. The average shortest path length stays almost unchanged, increasing only slightly from 4.00 to 4.03. This shows that even as the network expands, it does not become harder to navigate. Hubs continue to keep the network tightly connected, preserving the small-world structure.

*NOTE: The metrics for temporal evolution vary slightly between runs due to small calculation differences. However, these changes are not meaningful. For ex, the average path length may shift between 3.98 and 4.03, but it consistently stays around 4. This means the overall results are stable and reliable.*

## Sentiment Analysis

| Metric | Positive Network | Negative Network |
|---|---|---|
| **Nodes** | 35,010 | 6,370 |
| **Edges** | 130,513 | 14,513 |
| **Clustering Coefficient** | 0.1774 | 0.0945 |

Table 6. Sentiment Analysis

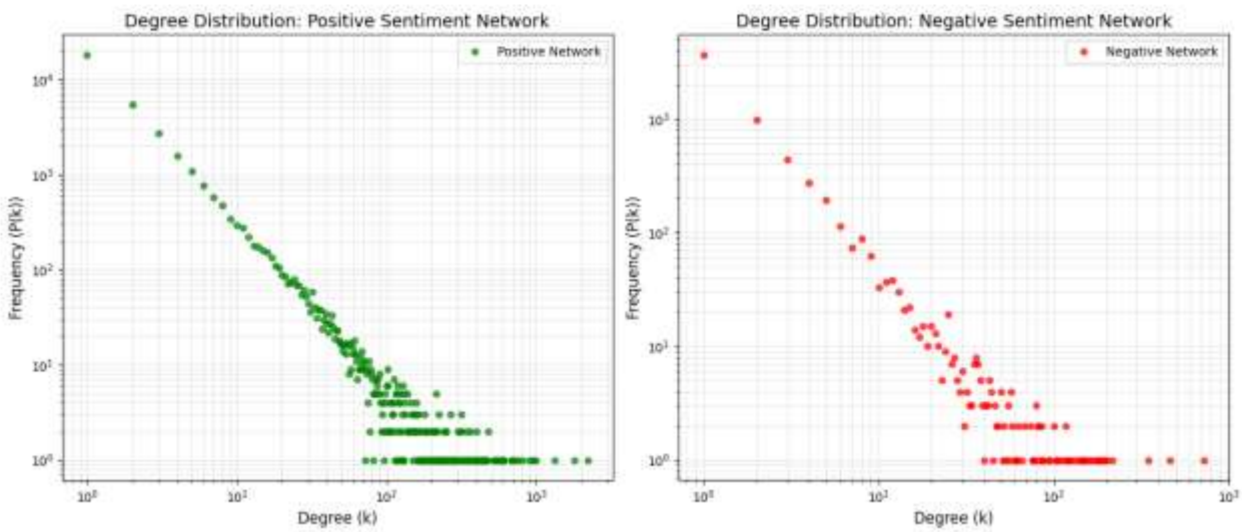


Figure 6. Sentiment Analysis

The dataset contained signed edges, meaning each interaction was labeled as either positive (+1) or negative (–1). In order to understand how supportive and hostile interactions shaped Reddit, I performed sentiment analysis which allowed me to compare how these two types of links behave structurally.

As displayed in Table 6, the positive network is much larger, with 35,010 nodes and 130,513 edges, compared to the negative network's 6,370 nodes and 14,513 edges. This matches the overall distribution of the data, where most interactions are positive or neutral. In other words, positive links form the main structure of the network, while negative links represent a smaller layer on top of it.

Structurally, both networks show heavy-tailed degree distributions, but they behave differently. The positive network contains many high-degree hubs and a long tail, meaning interaction is concentrated around major subreddits but still widely spread. The negative network has a much shorter tail, meaning fewer hubs and less concentration. Negative interactions are therefore more limited and less centered around the biggest communities.

Clustering also shows a clear difference. The positive network has a higher clustering coefficient (0.1774) than the negative network (0.0945), meaning positive interactions are more likely to form tight, interconnected groups. This suggests that positive links help reinforce community structure. In contrast, negative links are less clustered and more scattered, meaning they are less about building groups and more about crossing between them.

Overall, the positive network forms the core structure of Reddit, with dense, community-reinforcing connections. The negative network is smaller and more fragmented, and instead of building communities, it tends to highlight boundaries between them and often act as bridges.

## Community Detection

| Metric | Negative Network | Positive Network |
|---|---|---|
| **Nodes** | 6,370 | 35,010 |
| **Edges** | 14,019 | 117,594 |
| **Communities Detected** | 375 | 545 |
| **Largest Community Size** | 608 | 6,375 |
| **Modularity** | 0.5146 | 0.5200 |
| **Central Nodes Spread Across Communities** | High | Moderate |

Table 7. Community Detection

For community detection, the Louvain method was applied separately to the positive and negative sentiment networks to see how subreddits group based on interaction patterns. Both networks are strongly modular, but they differ in scale and cohesion.

The negative network is smaller (6,370 nodes, 14,019 edges) and forms 375 communities, with a largest community of 608 nodes. Modularity is 0.5146, which is strong but only slightly lower than the positive network. This indicates that while structure still exists, negative interactions are more broken up into smaller, less cohesive groups. In this network, subreddits like subredditdrama, askreddit, and iama act as key bridges, meaning negative interactions tend to pass through major hubs rather than staying within one group.

The positive network is much larger (35,010 nodes, 117,594 edges) and forms 545 communities, with a much larger main community of 6,375 nodes. Modularity is 0.5200, only slightly higher than the negative network. This small difference suggests both networks are similarly modular overall, but the positive one has more cohesive and larger community structures.

Overall, both networks are clearly community-driven, but positive interactions form larger, more stable groups, while negative interactions are more scattered across boundaries. Despite this difference, the same central subreddits appear in both, showing that conflict flows through the same core hubs that structure normal interactions rather than forming a separate network.

## Discussion

The goal of this project was to understand how conflict between subreddits shapes Reddit's overall structure. Across all analyses — degree, centrality, sentiment, community structure, and temporal evolution — a picture emerges: Reddit is a highly modular, hub-driven network where both positive and negative interactions flow through the same core structure.

**1. Do negative links increase separation between groups?**

The negative sentiment network is much smaller and less clustered (0.0945) than the positive network (0.1774), and its largest community is also far smaller (608 vs. 6,375 nodes). This illustrates that the negative interactions do not build strong internal structure. Instead, they are more fragmented and tend to appear at the edges of communities rather than reinforcing them.

**2. Are negative interactions mostly between communities?**

Yes. Negative betweenness results show that high-centrality nodes are spread across many different communities, not concentrated within one. This suggests that conflict tends to move across community boundaries rather than stay inside them. This matches prior findings where conflict often occurs between related but distinct communities rather than within them.

**3. Which subreddits connect the network?**

Across all centrality measures, the same core subreddits consistently dominate: *askreddit, iama, subredditdrama,etc.* As a result, these subreddits act as the structural backbone of Reddit and importantly, they also carry much of the negative interaction, meaning conflict flows through the same hubs that organize normal activity.

**4. Does the network change over time?**

Despite large growth in size, the structure remains stable. Clustering increases slightly, density decreases, and average path length stays around 4. This shows that Reddit becomes larger and more community-structured over time, but its overall small-world structure remains unchanged due to persistent hub connectivity.

## Conclusion

Overall, Reddit's structure is defined by a combination of strong community formation and a small number of highly connected hubs. Positive interactions reinforce dense, cohesive communities, while negative interactions are more scattered and tend to move between communities rather than within them. They are also more likely to act as bridges.

Most importantly, regarding conflict, it does not form a separate or isolated part of the network. Instead, it travels through the same central hubs that structure normal interaction. This means Reddit is not divided into "positive" and "negative" regions, instead both are layered on top of the same underlying network.

This aligns with prior work (Kumar et al., 2018) showing that online conflict is concentrated in a small number of highly central communities and spreads through existing structural pathways rather than emerging independently.

## Future Work

For future work, there are a couple of extensions that could improve this analysis. First, instead of just splitting the network into two time periods, breaking it into smaller windows like monthly, quarterly or annually would give a clearer view of how conflict and community structures change over time.

Second, adding text-based analysis of the actual posts could additionally help validate the sentiment links. This would make it easier to tell whether negative edges really represent conflict, or if they sometimes just reflect criticism or normal discussion.

## Limitations

A few limitations should be noted. First, due to computational constraints, methods like betweenness and closeness centrality were estimated using sampling and igraph's efficient implementations, which is necessary for a network of this scale.

Second, I couldn't fully visualize the entire network using the Kawai layout because of its size. I tried a sampling approach (10 random samples of up to 300 nodes), but even this doesn't capture the full structure. Sampling can hide key hubs, blur community boundaries, and miss important long-range connections between groups. Because of this, I did not include these graphs in the report, but they are available in the Colab notebook. These visualizations should be treated as a simplified view rather than the full network structure.

## Citations

S. Kumar, W.L. Hamilton, J. Leskovec, D. Jurafsky. Community Interaction and Conflict on the Web. World Wide Web Conference, 2018.